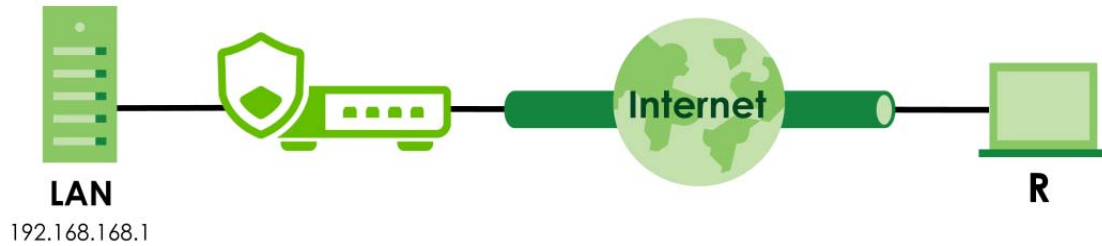


Figure 178 Remote Access VPN Example Topology



See the following table for VPN types and remote software options.

Table 119 SSL / IKEv2 VPN Comparison

VPN TYPE	FEATURE	HOME USER SOFTWARE	SUPPORTED OPERATING SYSTEMS	AUTHENTICATION REQUIREMENTS
SSL	Provides high security. May have lower connection speed and stability.	OpenVPN Connect	Windows, macOS, Linux, iOS, Android	- VPN account username and password - OVPN configuration file
		SecuExtender	Windows, macOS	VPN account username and password
IKEv2	Provides high security, connection speed and stability.	The IPSec VPN Client on Your Computer	Windows, macOS, iOS, and Android (strongSwan)	- VPN account username and password - Configuration file
		SecuExtender	Windows, macOS	VPN account username and password

13.5.1 Zyxel Device Setup

Select [SSL](#) or [IKEv2](#) to configure the Zyxel Device in the office. See [Table 119 on page 250](#) for the comparison between two VPN types.

SSL

- 1 Go to User & Authentication > User/Group > User, and click Add under User to create a VPN user account.

User					
+ Add Edit Remove Reference					
☐ Name	User Type	Description	Created Date	Password Changed Date	Reference
☐ zyxel_vpn	user		2024-11-01 14:10	2024-11-01 14:10	2
☐ radius-users	ext-user		Built-in	-	0
☐ ldap-users	ext-user		Built-in	-	0
☐ ad-users	ext-user		Built-in	-	0

- 2 Set a VPN user name and password, then click Apply to save your changes. Note down the account name and password for the home user who will use this for future remote access authentication.

← User & Authentication > User/Group > User

Profile Management

User Name	zyxel_vpn
User Type	User
Password	
Retype	
Description	
Email 1	
Email 2	
Mobile Number	

Authentication Timeout Settings

☒ Use Default Settings ☐ Use Manual Settings

Lease Time	1440	minutes
Reauthentication Time	1440	minutes

Two-factor Authentication

Enable Two-Factor Authentication for VPN Access ☐

Some changes were made

What do you want to do then?

Cancel Apply

- 3 To configure SSL VPN on the Zyxel Device, go to VPN > SSL VPN.

VPN > SSL VPN

General Settings

Zyxel Remote VPN works with the SecuExtender VPN client and is also compatible with the OpenVPN Connect client.

Enable ☒ ⓘ

SSL VPN Configuration Download [Download](#)

Incoming Interface

Interface:

DNS Name: (Optional)

Server Port:

Zone: ⓘ

Clients will use VPN to access

☒ Internet and Local Networks (Full Tunnel)

Auto SNAT ☒ ⓘ

☐ Local Networks Only (Split Tunnel)

Client Network

IP Address Pool:

First DNS Server: ☒ ZyWALL

☐ Custom Defined

Second DNS Server:

Authentication ⓘ

Primary Server:

Secondary Server:

User: ⓘ

[Advanced Settings](#) ▾

Some changes were made

What do you want to do then?

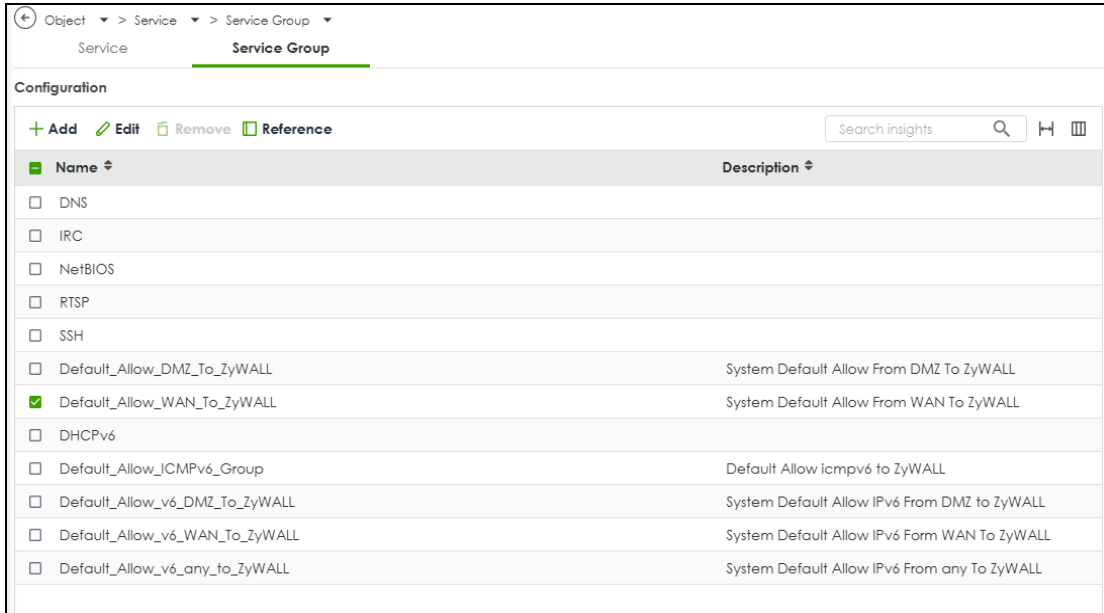
[Cancel](#) [Apply](#)

Follow the table below to configure the VPN > SSL VPN screen.

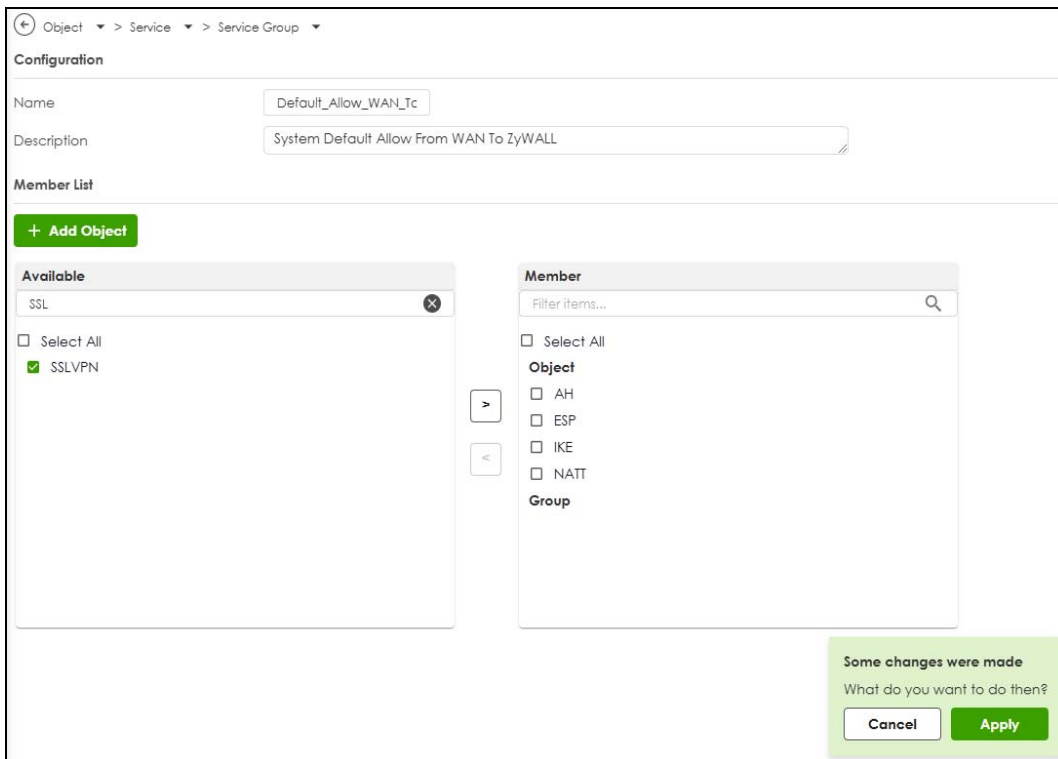
Table 120 SSL VPN Screen Configuration

LABEL	DESCRIPTION
Enable	Click this to the right to enable SSL VPN.
Interface	Select an interface for incoming traffic to your Zyxel Device.
Clients will use VPN to access	Full Tunnel - Select this to encrypt all traffic through the VPN. Split Tunnel - Select this to only encrypt traffic going to networks behind the Zyxel Device. Enter an IPv4 address in CIDR notation, for example, type IP address 192.168.51.0/24. Traffic going to the Internet from this IP address is encrypted, and not encrypt traffic going to the Internet through the Zyxel Device.
User	Select the user account you created in step 2 to allow SSL VPN access

- Click Apply to save the changes.
- To allow the Zyxel Device to access VPN traffic from WAN, go to Object > Service > Service Group. Select Default_Allow_WAN_To_ZyWALL and click Edit.



- 6 Search for SSL VPN under Available and click > to add it to the allow list of traffic from the WAN to the Zyxel Device. Then, click Apply to save the changes.



IKEv2

- 1 Go to User & Authentication > User/Group > User, and click Add under User to create a VPN user account.

User					
+ Add Edit Remove Reference					
☐	Name	User Type	Description	Created Date	Password Changed Date
☐	zyxel_vpn	user		2024-11-01 14:10	2024-11-01 14:10
☐	radius-users	ext-user	Built-in	-	0
☐	ldap-users	ext-user	Built-in	-	0
☐	ad-users	ext-user	Built-in	-	0

- Set a VPN user name and password, then click Apply to save your changes. Note down the account name and password for future remote access authentication.

User & Authentication > User/Group > User

Profile Management

User Name

zyxel_vpn

User Type

User

Password

....

Retype

....

Description

Email 1

Email 2

Mobile Number

Authentication Timeout Settings

☒ Use Default Settings
☐ Use Manual Settings

Lease Time

1440

minutes

Reauthentication Time

1440

minutes

Two-factor Authentication

Enable Two-Factor Authentication for VPN Access

☐

Some changes were made

What do you want to do then?

Cancel

Apply

- To configure IKEv2 VPN on the Zyxel Device, go to VPN > IPSec VPN > Remote Access VPN and enable IKEv2 VPN.

VPN > IPsec VPN > Remote Access VPN

Site to Site VPN **Remote Access VPN**

General Settings

ZyXel's remote VPN solution uses leading IPsec/IKEv2 (EAP-MSCHAPv2) encryption, supported by SecuExtender VPN Client. You can also use native clients built into Windows, Android, macOS and iOS.

Enable ☒

Get SecuExtender VPN Client Software [Windows](#) [macOS](#)

VPN Configuration Download for Native VPN Client [Windows](#) [iOS/macOS](#) [Android \(strongSwan\)](#)

Incoming Interface

☒ Interface

☐ Domain Name / IP

NAT Traversal

Zone

Certificate for VPN Validation

☒ Auto

☐ Manual

Clients will use VPN to access

☐ Internet and Local Networks (Full Tunnel)

Auto SNAT ☒

☒ Local Networks Only (Split Tunnel)

Local Network

Client Network

IP Address Pool

First DNS Server ☒ ZyWALL

☐ Custom Defined

Second DNS Server

Authentication

Primary Server

Secondary Server

User

Some changes were made
What do you want to do then?
[Cancel](#) [Apply](#)

Follow the table below to configure the VPN > IPsec VPN > Remote Access VPN screen.

Table 121 IKEv2 VPN Screen Configuration

LABEL	DESCRIPTION
Enable	Click this to the right to enable SSL VPN.
Interface	Select an interface for incoming traffic to your Zyxel Device.
Clients will use VPN to access	Internet and Local Networks (Full Tunnel) - Select this to encrypt all traffic through the VPN. Local Networks Only (Split Tunnel) - Select this to only encrypt traffic going to networks behind the Zyxel Device. Enter an IPv4 address in CIDR notation, for example, type IP address 192.168.51.0/24. Traffic going to the Internet from this IP address is encrypted, and not encrypt traffic going to the Internet through the Zyxel Device.
User	Select the user account you created in step 2 to allow IKEv2 VPN access

- Click Apply to save your changes.
- Send authentication details to the home user.

13.5.2 Home User Setup

The administrator has now finished setting up the VPN configuration on the Zyxel Device. Now, the home user needs to set up a VPN client software on their computer or mobile device to connect to the office network. See [Table 119 on page 250](#) for VPN software options for home user and more details.

SecuExtender

SecuExtender is a Zyxel subscription-based VPN client.

Home users using SecuExtender need the following:

- The SecuExtender VPN client software: They should get this from the Zyxel Device administrator, who downloads it from the VPN > IPSec VPN > Remote Access VPN > Get SecuExtender VPN Client Software screen. Alternatively, you can download it directly from the Zyxel website.
- VPN account username and password: They should get this from the Zyxel Device administrator, who sets it in the User & Authentication > User/Group > User screen.

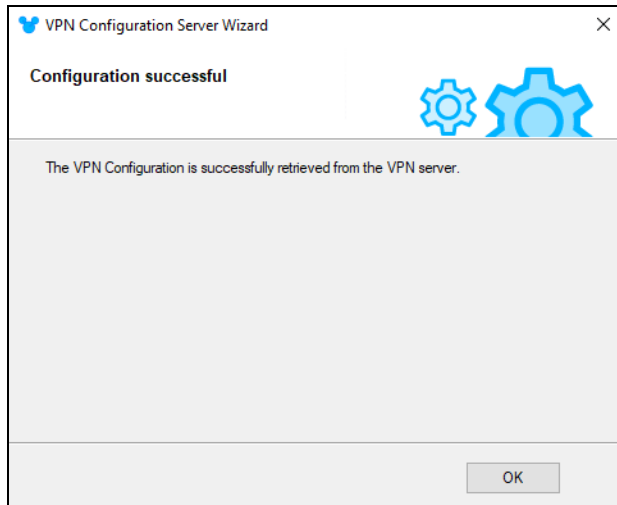
Follow these steps to establish a VPN connection to the office's network through SecuExtender:

- 1 Unzip, install, and open the SecuExtender VPN Client on your computer. Click Configuration > Get from Server, then enter the parameters as described below and click Next.

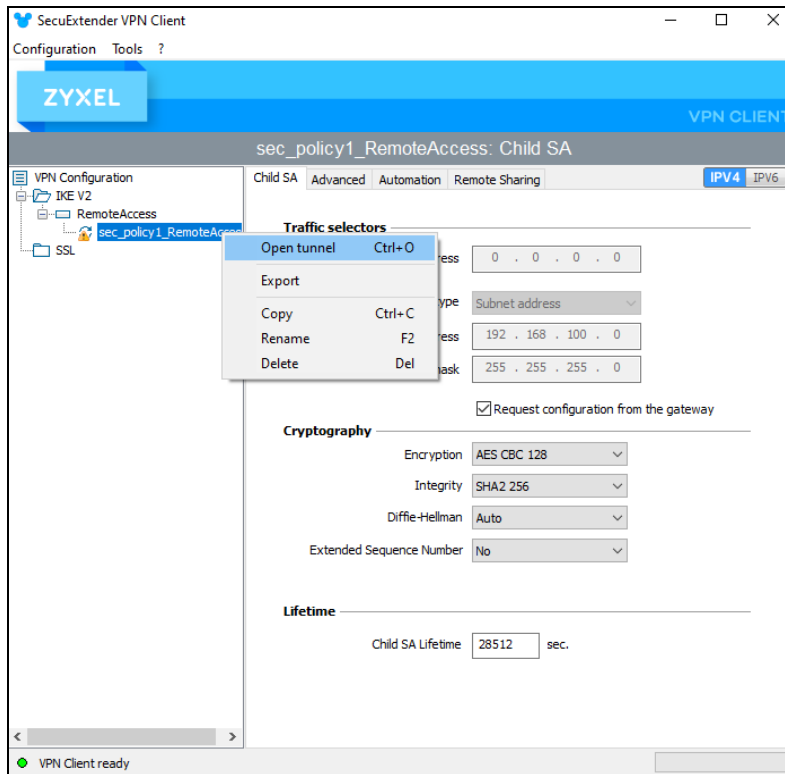
Note: You must log in with an account of type User, not Local Administrator.

LABEL	DESCRIPTION
Gateway Address	Enter the WAN IP address of the Zyxel Device.
Authentication	Set as Login + Password.
Login/Password	Enter the username and password the Zyxel Device administrator gave.

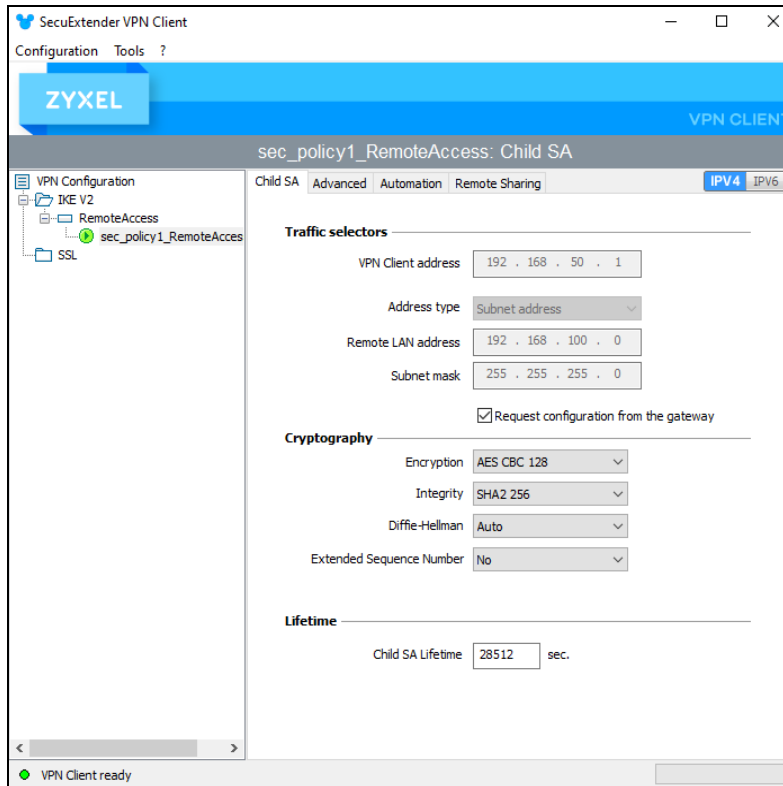
- 2 The following screen appears, click OK.



- 3 Right click on the VPN policy you just created, then click Open tunnel to establish a remote VPN connection.



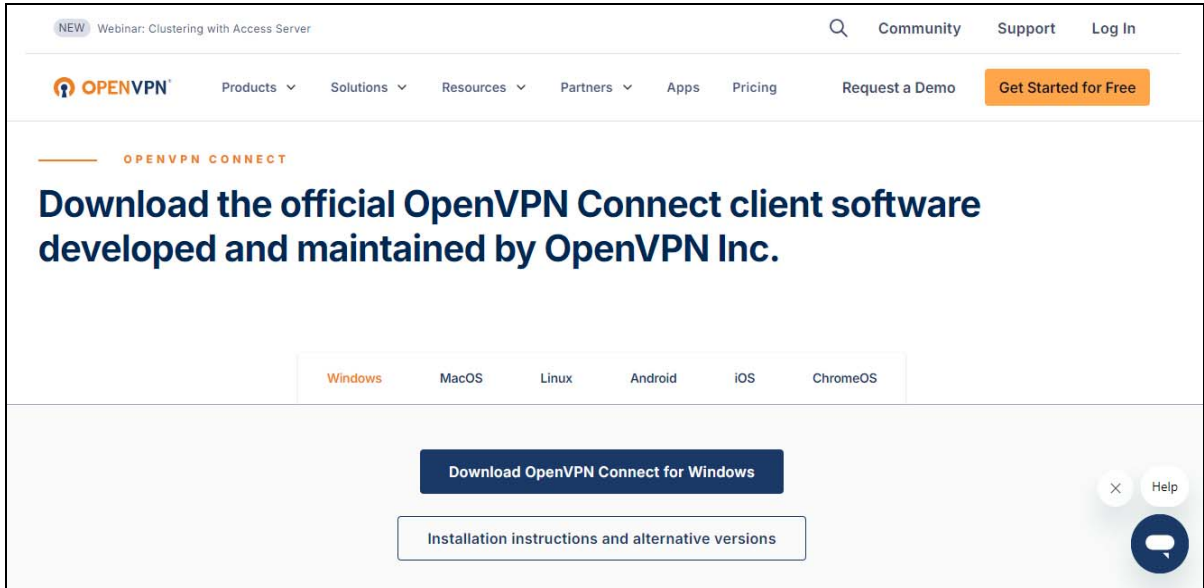
- 4 Re-enter the user name and password, then click OK. The icon next to the VPN policy turns green. You can now access the office network through the Zyxel Device.



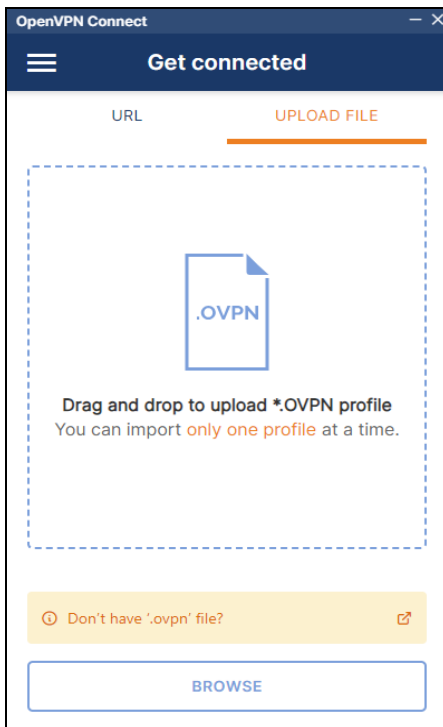
OpenVPN Connect

Follow these steps to establish a VPN connection to the office's network through OpenVPN Connect:

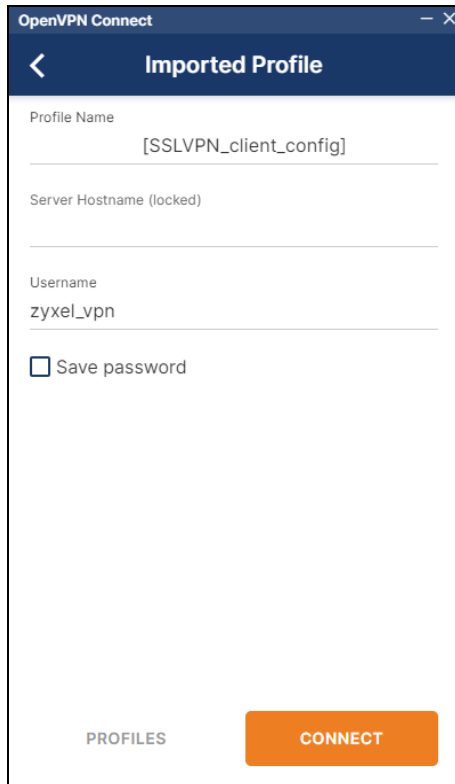
- 1 Home users using OpenVPN Connect need the following:
 - The OpenVPN Connect client software.
 - The VPN account username and password: They should get this from the Zyxel Device administrator, who sets it in the User & Authentication > User/Group > User screen.
 - The OVPN configuration file: They should get this from the Zyxel Device administrator, who downloads it from the VPN > SSL VPN screen.
- 2 Go to the [OpenVPN Connect](#) website and download the OpenVPN Connect client for your computer's operation system.



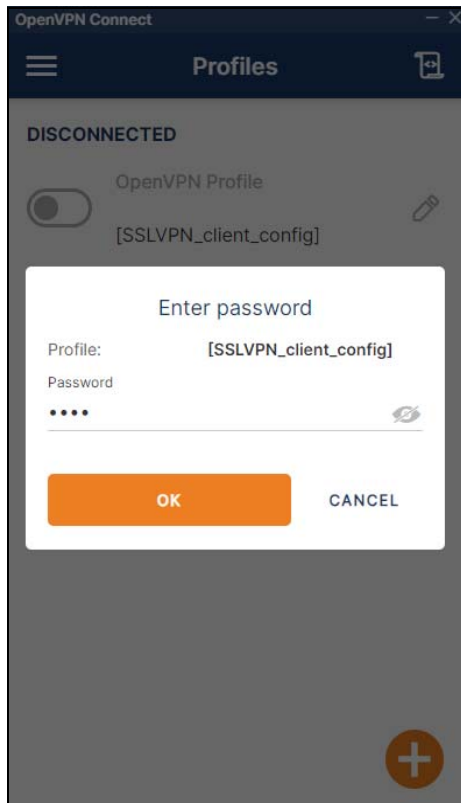
- 3 Run the OpenVPN Connect client on your computer. Click Browse and import the .OVPN file provided by Zyxel Device administrator.



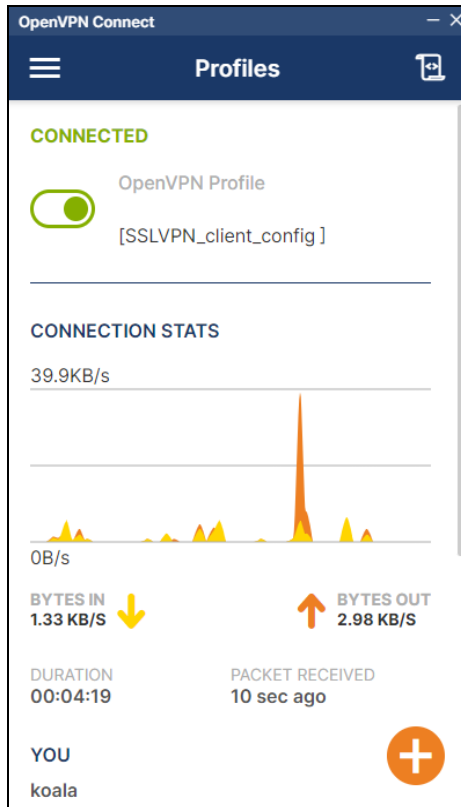
- 4 In the Username field, enter the VPN user name the Zyxel Device administrator set. Click Connect to connect your computer to the office network.



- 5 Enter the VPN user password provided by the Zyxel Device administrator.



- 6 Your home computer can now access the office network through the Zyxel Device.



The IPSec VPN Client on Your Computer

Use the built-in VPN client in Windows, macOS, iOS, or Android (strongSwan).

Home users using the IPSec VPN client on their computers need the following:

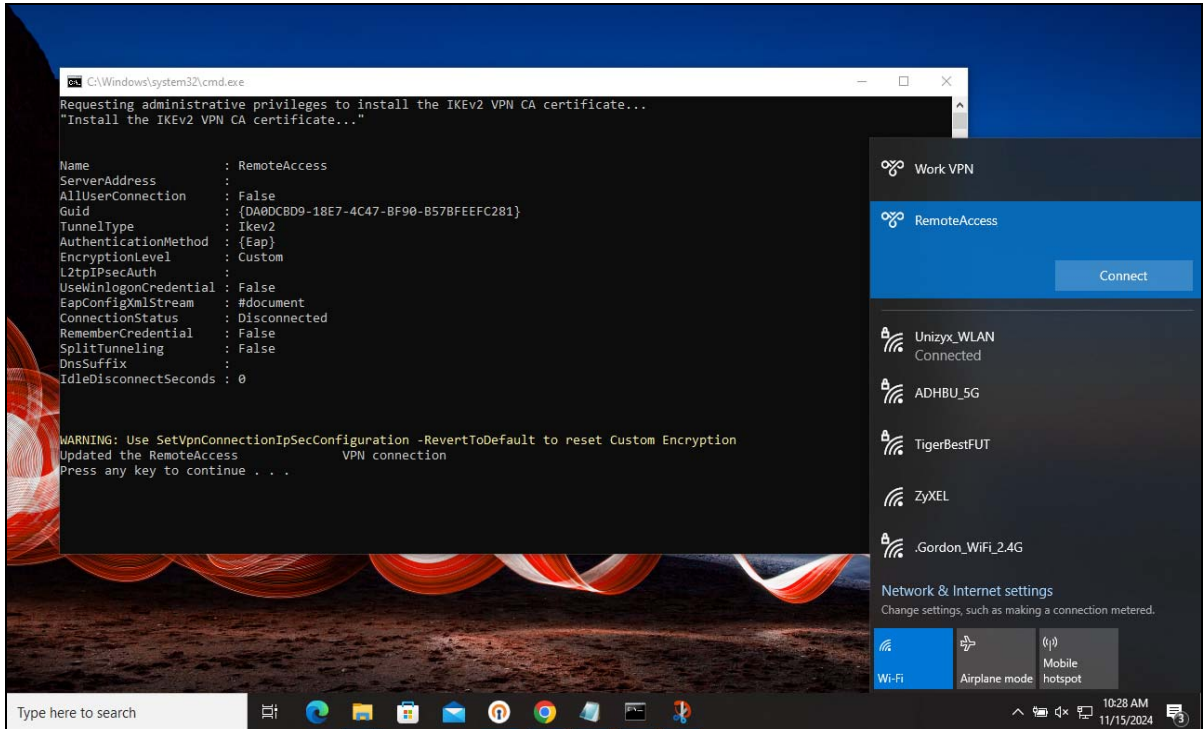
- Configuration file: They should get this from the Zyxel Device administrator, who downloaded it from the VPN > IPSec VPN > Remote Access VPN screen.

Follow these steps to establish a VPN connection to the office's network through the IPSec VPN client on a computer with a Windows operating system:

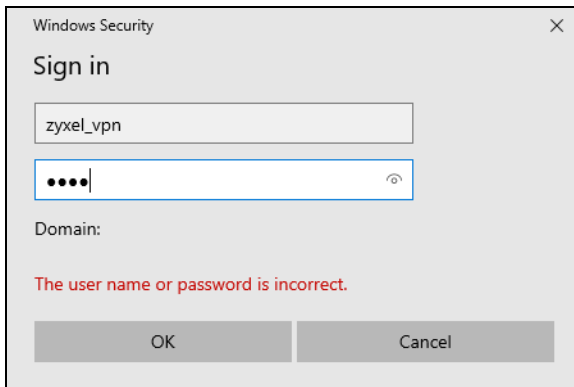
- 1 Obtain the configuration file, VPN account name and password from the Zyxel Device administrator.
- 2 Unzip and open the configuration file, then double-click on the .bat file to set up the certificate for the VPN connection.

Name	Date modified	Type	Size
Readme	14/11/2024 18:26	Text Document	1 KB
RemoteAccess_Win_RemoteAccess	14/11/2024 18:26	Windows Batch File	3 KB
RemoteAccess_Win_RemoteAccess	14/11/2024 18:26	Security Certificate	1 KB

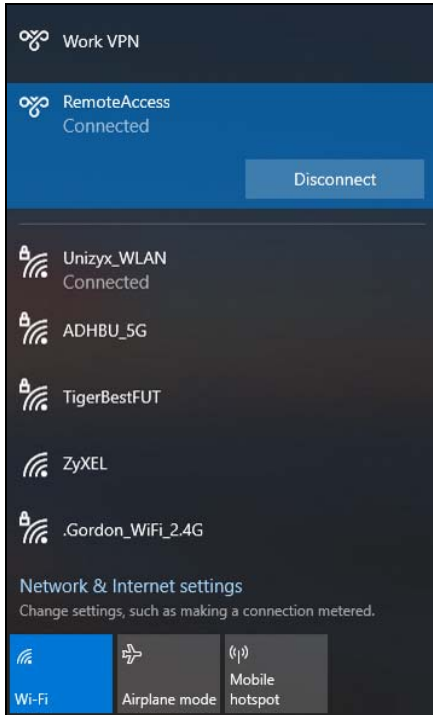
- 3 A command-line interface will appear, showing the status of the VPN connection. To connect to the office network, click the Internet access icon, then click Connect next to the RemoteAccess network.



- 4 Enter the username and password provided by the administrator in the pop-up window, then click OK.



- 5 The following screen indicates you are now connected to the office network.



13.5.3 Test the VPN Connection

To test if the home user's computer can successfully connect to the office's network, they should open the Command Prompt and ping the IP address of a device in the LAN. If the connection is successful, the following result will appear.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5073]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NT03315>ping 192.168.168.1

Pinging 192.168.168.1 with 32 bytes of data:
Reply from 192.168.168.1: bytes=32 time<1ms TTL=64
Reply from 192.168.168.1: bytes=32 time<1ms TTL=64
Reply from 192.168.168.1: bytes=32 time<1ms TTL=64
Reply from 192.168.168.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.168.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\NT03315>
```